

RAPHAEL ESQUIVEL

Email: esquivelralphie@gmail.com

EDUCATION

Harvey Mudd College, Claremont, CA

Aug 2025 - May 2029

B.S., Mathematics & Physics

*Related courses: Readings in Algebraic Geometry / Numerical Analysis / Abstract Algebra 1
/ Intermediate Probability / Quantum Field Theory*

Arizona State University, Phoenix, AZ

Aug 2024 - Apr 2025

Non-Degree Seeking Student (GPA: 4.0)

*Related courses: Mathematical Methods Physics II / Introductory Modern Physics
/ Quantum Physics II / Classical Particles, Fields, and Matter II*

EXPERIENCE

Student Researcher, Quantum Quasars Lab (HMC)

Sep 2025 - Jul 2026

Designing, assembling, and calibrating free-space quantum communications systems under Prof. Jason Gallicchio. Worked on entanglement source, quantum link, Pockels cell.

Student Researcher, Logarithmic Gromov-Witten Theory (HMC)

Nov 2025 - May 2026

Analyzed the tropical and combinatorial structure of relative stable maps to $(\mathbb{P}^n, H)$ and their boundary behavior in moduli spaces under Prof. Dagan Karp.

AWARDS

Karl Menger Memorial Award (2nd), American Mathematical Society

May 2025

Awarded \$1000 at Regeneron ISEF for project *Lie Groups and Algebroids in Lagrangian Reduction*.

SKILLS

Software

Python, Javascript, C/C++, R, Mathematica, Julia, 3D modeling

Hardware

FDM, SLS, lathe, mill, 3D printer maintenance

Languages

English (Native), French (Native), Spanish (Proficient)

Research interests

Enumerative geometry, logarithmic Gromov-Witten theory, tropical algebraic geometry, topological quantum field theory, geometric mechanics